

# Axler Linear Algebra

Nathan Yan

July 19, 2026

## Contents

|     |                                             |    |
|-----|---------------------------------------------|----|
| 1   | Vector Spaces                               | 3  |
| 1.A | $\mathbb{R}^n$ and $\mathbb{C}^n$ . . . . . | 3  |
| 1.B | Definition of a Vector Space . . . . .      | 4  |
| 1.C | Subspaces . . . . .                         | 7  |
| 2   | Finite Dimensional Vector Spaces            | 13 |
| 2.A | Span and Linear Dependence . . . . .        | 13 |
| 2.B | Bases . . . . .                             | 16 |
| 2.C | Dimension . . . . .                         | 18 |
| 3   | Linear Maps                                 | 23 |
| 3.A | Vector Space of Linear Maps . . . . .       | 23 |
| 3.B | Null Spaces and Ranges . . . . .            | 27 |

I plan to mostly take notes on paper, but I will a lot of the exercises here because I need to practice writing  $\text{\LaTeX}$  quickly and stuff.

# 1 Vector Spaces

## 1.A $\mathbb{R}^n$ and $\mathbb{C}^n$

These exercises are really easy but I am always scared that my proofs of these are not rigorous enough....

I do odds.

1 Show that  $\alpha + \beta = \beta + \alpha$  for all  $\alpha, \beta \in \mathbb{C}$ .

Proof. Let  $\alpha = a + bi$  and  $\beta = c + di$  where  $a, b, c, d \in \mathbb{R}$ .

Then we have

$$\begin{aligned}\alpha + \beta &= \\ a + bi + c + di &= \\ (a + c) + (b + d)i &= \end{aligned}$$

Because addition is commutative under the reals, we can swap the terms.

$$\begin{aligned}(c + a) + (d + b)i &= \\ c + di + a + bi &= \\ \beta + \alpha & \end{aligned}$$

□

3 Show that  $(\alpha\beta)\lambda = \alpha(\beta\lambda)$  for all  $\alpha, \beta, \lambda \in \mathbb{C}$ .

Distribute, multiply, then rearrange terms using the commutativity of the reals to get the answer. It is quite tedious, so I won't do it here.

5 Show that for every  $\alpha \in \mathbb{C}$ , there exists a unique  $\beta \in \mathbb{C}$  such that  $\alpha + \beta = 0$ .

Proof. Let  $\alpha = a + bi$ . Then, let  $\beta = -a - bi$ . We can observe that

$$\begin{aligned}\alpha + \beta &= \\ a + bi - a - bi &= \\ (a - a) + i(b - b) &= 0 \end{aligned}$$

□

7 Show that  $\frac{-1+\sqrt{3}i}{2}$  is a cube root of 1 (meaning that its cube equals 1)

Again, this is a tedious problem where you just multiply it out. Of course, I can represent this number as  $e^{\frac{\pi}{3}i}$  but that's cheating.

9 Find  $x \in \mathbb{R}^4$  such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$$

Proof. This is vector addition, so we can apply the definition.

$$\begin{aligned}(4, -3, 1, 7) + 2x &= (5, 9, -6, 8) \\ (4 + 2x_1, -3 + 2x_2, 1 + 2x_3, 7 + 2x_4) &= (5, 9, -6, 8)\end{aligned}$$

For each term in the list, the number must hold, so in essence we get four different equations:

$$\begin{aligned}4 + 2x_1 &= 5 \\ -3 + 2x_2 &= 9 \\ 1 + 2x_3 &= -6 \\ 7 + 2x_4 &= 8\end{aligned}$$

Each of these linear equations yields only one solution, so the only solution for the equation is  $x = (\frac{1}{2}, -\frac{3}{2}, -\frac{7}{2}, \frac{1}{2})$  □

## 1.B Definition of a Vector Space

This one introduces the concept of a vector space in an abstract manner and gives a few unorthodox examples.

1 Prove that  $-(-v) = v$  for every  $v \in V$

Proof.  $-(-v)$  is the additive inverse of  $-v$ . Therefore,

$$\begin{aligned}-(-v) + (-v) &= 0 \\ -(-v) + (-v + v) &= v \\ -(-v) + 0 &= v \\ -(-v) &= v\end{aligned}$$

As desired. □

3 Suppose  $v, w \in V$ . Explain why there exists a unique  $x \in V$  such that  $v + 3x = w$ .

$$\begin{aligned}v + 3x &= w \\ 3x &= w - v \\ x &= \frac{1}{3}(w - v)\end{aligned}$$

Since that right side is a unique vector,  $x$  is a unique vector. You invoke the properties and stuff for this, but I'm too lazy.

- 5 Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of  $V$ .

Proof. We can show that given this condition, every element  $v \in V$  has an additive inverse.

$$\begin{aligned} 0v &= 0 \\ (1 + -1)v &= 0 \end{aligned}$$

Note that even though we don't know whether vector addition has an additive inverse the 0 on the left side is just a scalar, so we can still say there is an additive inverse.

Then distribute and apply the multiplicative identity axiom...

$$\begin{aligned} 1v + (-1)v &= 0 \\ v + (-1)v &= 0 \end{aligned}$$

Therefore  $(-1)v$  is the additive inverse for all  $v \in V$ . □

- 7 Suppose  $S$  is a nonempty set. Let  $V^S$  denote the set of functions from  $S$  to  $V$ . Define a natural addition and scalar multiplication on  $V^S$ , and show that  $V^S$  is a vector space with these definitions.

Proof. Like with  $F^S$ , we can sort of “piggyback” off the properties of  $V$  to maintain the properties of a vector space in  $V^S$ .

Define addition as

$$(f + g)(x) = f(x) + g(x)$$

Let the additive identity be  $0(x) = 0$  for all  $x \in S$ .

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x)$$

For all  $f, g \in V^S$ . Because  $f(x)$  is a vector, it has an inverse (namely  $-f(x)$ ). Therefore, we can define the additive inverse as

$$(-f)(x) = -f(x)$$

$$(f + (-f))(x) = f(x) + (-f(x)) = 0 = 0(x)$$

Commutativity:

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$$

We can swap  $f(x)$  and  $g(x)$  because  $V$  is commutative.

Associativity:

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$$

Again, we use the properties of the vector space  $V$  to help us out here.

Scalar multiplication

For all  $\lambda \in \mathbb{F}, v \in V^S$ ,

$$(\lambda v)(x) = \lambda v(x)$$

Multiplicative Identity

1 is the multiplicative identity.

$$(1v)(x) = 1v(x) = v(x)$$

Since 1 is the multiplicative identity of  $V$ .

Distributive property

$$(a(u + v))(x) = a(u + v)(x) = a(u(x) + v(x)) = au(x) + av(x) = (au + av)(x)$$

Because of the distributive property of  $V$ .

$$((a + b)v)(x) = (a + b)v(x) = av(x) + bv(x) = (av + bv)(x)$$

For the same reason.

Closure

Because  $V$  is closed under addition and scalar multiplication, and all our operations are defined using these 2 operations,  $V^S$  is closed.

Thus, we have proved all the properties of vector spaces on  $V^S$ , therefore  $V^S$  is a vector space.

□

## 1.C Subspaces

This is where it ramps up a little bit, we are doing some linear algebra now! This idea of “subspaces” is interesting...

1 For each of the following subsets of  $\mathbb{F}^3$ , determine whether it is a subspace of  $\mathbb{F}^3$

(a)  $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$

Proof.  $(0, 0, 0)$  is in our set, so we have the additive identity. Consider vectors  $x$  and  $y$  in our set, and add them.

$$\begin{aligned}x + y &= \\(x_1, x_2, x_3) + (y_1, y_2, y_3) &= \\(x_1 + y_1, x_2 + y_2, x_3 + y_3) &= \\z &= \\z_1 + 2z_2 + 3z_3 &= \\x_1 + y_1 + 2x_2 + 2y_2 + 3x_3 + 3y_3 &= \\x_1 + 2x_2 + 3x_3 + y_1 + 2y_2 + 3y_3 &= \\0 + 0 &= 0\end{aligned}$$

Because  $x$  and  $y$  are members of the subset, we can use the condition. Therefore,  $z$  is part of our set.

Now let us consider scalar multiplication. Consider  $x$  in our set and  $\lambda \in \mathbb{F}$ . Then,

$$\begin{aligned}\lambda x &= \\(\lambda x_1, \lambda x_2, \lambda x_3)\end{aligned}$$

We then consider whether this new element is in our set.

$$\begin{aligned}\lambda x_1 + 2\lambda x_2 + 3\lambda x_3 &= \\\lambda(x_1 + 2x_2 + 3x_3) &= \\\lambda(0) &= 0\end{aligned}$$

We use the distributive property here. Since it follows the set's condition, we can say that it is closed under scalar multiplication.

These are all the conditions, so we can say that our set is a subspace of  $\mathbb{F}^3$ .  $\square$

(b)  $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$

Proof. Consider the element  $(1, 0, 1)$  which is in our set because  $1 + 2 \cdot 0 + 3 \cdot 1 = 4$  and it is in  $\mathbb{F}^3$ . Now consider multiplying it by the scalar 2 to create the element  $(2, 0, 2)$ . Unfortunately,  $2 + 2 \cdot 0 + 3 \cdot 2 = 8 \neq 4$ , so this set does not fulfill the closure under scalar multiplication condition.  $\square$

(c)  $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$

Proof. Consider the sets  $(1, 1, 0)$  and  $(0, 0, 1)$ . Summing these up, we get  $(1, 1, 1)$ , but  $1 \cdot 1 \cdot 1 = 1 \neq 0$ , so this set fails the additive closure property.  $\square$

(d)  $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

I am too lazy but I did the other 3, good enough

2 I did number 2 in my notebook, not noticing it was actually an exercise.

3 Show that the set of differentiable real-valued functions  $f$  on the interval  $(-4, 4)$  such that  $f'(-1) = 3f(2)$  is a subspace of  $\mathbf{R}^{(-4,4)}$ .

Proof. Clearly this set is a subset of  $\mathbb{R}^{(-4,4)}$ . The zero function ( $f(x) = 0$ ) satisfies the condition (since the derivative is also 0), so we have the additive identity.

Consider functions  $f, g$  in the set, then consider  $(f + g)(x)$ . Functions and their derivatives are additive, so we can write

$$\begin{aligned}(f + g)'(-1) &= \\ f'(-1) + g'(-1) &= \\ 3f(2) + 3g(2) &= \\ 3(f(2) + g(2)) &= \\ 3(f + g)(2) &= \end{aligned}$$

Therefore, we have additive closure.

Consider  $f$  in the set and  $\lambda \in \mathbb{R}$ . Let us show that  $(\lambda f)(x) = \lambda f(x)$  is in the set.

$$\begin{aligned}(\lambda f)'(-1) &= \\ \lambda f'(-1) &= \\ 3\lambda f(2) &= \\ 3(\lambda f)(2) &= \end{aligned}$$

So we are closed under scalar multiplication.  $\square$

5 Is  $\mathbb{R}^2$  a subspace of the complex vector space  $\mathbb{C}^2$ ?

Proof. Although  $\mathbb{R}^2$  is a vector space over  $\mathbb{R}$ , it is not a vector space over  $\mathbb{C}$ . For example, multiplying  $(1, 1)$  by  $i$  produces a vector not in  $\mathbb{R}^2$ , so it is not closed under scalar multiplication.  $\square$

7 If  $U$  is closed under addition and under taking additive inverses (meaning  $-u \in U$  whenever  $u \in U$ ), then  $U$  is a subspace of  $\mathbb{R}^2$ .

Proof. Consider the set  $\mathbb{Q}^2$ . It is closed under addition and additive inverse, but it is not closed under scalar multiplication over  $\mathbb{R}$ . For example, the set  $(1, 1)$  when multiplied by an irrational number such as  $\sqrt{2}$  is not a part of the set.  $\square$

- 9 A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called periodic if there exists a positive number  $p$  such that  $f(x) = f(x + p)$  for all  $x \in \mathbb{R}$ . Is the set of periodic functions from  $\mathbb{R}$  to  $\mathbb{R}$  a subspace of  $\mathbb{R}^{\mathbb{R}}$ ? Explain.

Proof. There is not closure under addition, unfortunately. Consider the functions

$$\begin{aligned} f(x) &= \cos(2\pi x) \\ g(x) &= \cos(2x) \end{aligned}$$

$f(x)$  is at its maximum of 1 whenever  $x \in \mathbb{Z}$ , and  $g(x)$  is at its maximum of 1 whenever  $x = n\pi, n \in \mathbb{Z}$ .  $(f + g)(x) = 2$  only when both of these conditions are met. However, this is only true when  $x = 0$ .

Suppose that  $x = n\pi$  and  $x = m$  where  $n, m \in \mathbb{Z}$ . Then,

$$\begin{aligned} n\pi &= m \\ \pi &= \frac{m}{n} \end{aligned}$$

Which implies that  $\pi$  is rational if  $n \neq 0$ , so the only solution is  $x = 0$ .

Therefore,  $(f + g)(x)$  reaches its maximum only at  $x = 0$ , and there is no  $p > 0$  where  $(f + g)(0) = (f + g)(p)$ . Therefore  $(f + g)(x)$  is outside the set and it fails the additive closure condition, making it not a vector space.  $\square$

- 11 Prove that the intersection of every collection of subspaces of  $V$  is a subspace of  $V$ .

Proof. Consider subspaces of  $V$ ,  $V_1, \dots, V_M$ . All subspaces contain the identity element, so  $V_1 \cap \dots \cap V_M$  will also contain it. Consider  $u, v \in V_1 \cap \dots \cap V_M$ . Because  $u, v \in V_k$ , any addition and scalar multiplication will remain in  $V_k$ . Therefore, any addition and scalar multiplication will remain in  $V_1 \cap \dots \cap V_M$ , completing the proof.  $\square$

- 12 Prove that the union of two subspaces of  $V$  is a subspace of  $V$  if and only if one of the subspaces is contained in the other.

First, let's prove a lemma we will need for this.

Lemma. Given  $u \notin V, v \in V, u + v \notin V$ .

Proof. Suppose that  $u + v \in V$ . Then, because of the additive inverse,  $u + v - v = u \in V$ , and we reach a contradiction.  $\square$

Now we can prove the main result.

Proof. Consider subspaces  $V_1, V_2$

Consider WLOG  $V_1 \subseteq V_2$ . Then,  $V_1 \cup V_2 = V_2$ , and since  $V_2$  is a subspace so is  $V_1 \cup V_2$ .

The converse is trickier. We prove the contrapositive. Consider when  $V_1$  and  $V_2$  are not subsets of each other. Then, there is some element  $u \in V_1, u \notin V_2$ . There must also

be an element  $v \in V_2, v \notin V_1$ . These are true from the definition of subset. Consider  $u + v$ . By the lemma, this sum is in neither set, so it is not in the union of  $V_1$  and  $V_2$ , so  $V_1 \cup V_2$  is not a subspace. □

- 13 Prove that the union of three subspaces of  $V$  is a subspace of  $V$  if and only if one of the subspaces contains the other two.

This problem is very difficult (as the textbook alluded) and I needed to look up the solution.

Proof. If any subspace is contained within another this reduces to the 2-subspace case. Presume that  $V_1 \cup V_2 \cup V_3$  is a subspace.

Suppose  $V_1 \cup V_2 \setminus V_3$  was not empty. Then, we can use  $u \in V_1 \cup V_2 \setminus V_3$  and  $v \in V_3 \setminus V_2 \cup V_1$  to prove that there is not closure using the lemma from exercise 12. By symmetry, there is no element that is in exactly two of the subsets.

From here on, suppose  $u \in V_1 \setminus V_2 \cup V_3, v \in V_2 \setminus V_1 \cup V_3$ . Note that if these elements did not exist one set would be a subset of another or we would have an element in exactly two subsets. Then, to be closed,  $u + v \in V_3$  since by the Exercise 12 lemma we cannot have  $u + v$  be in either  $V_1$  or  $V_2$ . Consider  $a, b$  in  $\mathbb{F}$  where  $a, b \neq 0, a - b \neq 0$ . We also have  $au + v \in V_3$  and  $bu + v \in V_3$  since scalar multiplication by a non-zero constant should not change which subspaces a vector is in, so  $au + v - (bu + v) = (a - b)u \in V_3$ . However,  $u$  was explicitly only in  $V_1$ , so we have a contradiction and  $V_1 \cup V_2 \cup V_3$  is not actually a subspace! Note that this proof does not work in  $\mathbb{F}_2$  because we do not have 2 different numbers that satisfy the properties of  $a$  and  $b$ . □

- 15 Suppose  $U$  is a subspace of  $V$ . What is  $U + U$ ?

Proof. We use the definition of subspace addition.

$$U + U = \{u + v : u, v \in U\}$$

Because  $U$  is a subspace and is therefore closed, this is identical to  $U$ . □

- 17 Is the operation of addition on the subspaces of  $V$  associative? In other words, if  $V_1, V_2, V_3$  are subspaces of  $V$ , is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Proof. Using the definition of subspace, we have

$$\begin{aligned} (V_1 + V_2) + V_3 &= \{(v_1 + v_2) + v_3 : v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= \{v_1 + (v_2 + v_3) : v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= V_1 + (V_2 + V_3) \end{aligned}$$

using the associative property of vector addition. □

19 Prove or give a counterexample: If  $V_1, V_2, U$  are subspaces of  $V$  such that

$$V_1 + U = V_2 + U$$

then  $V_1 = V_2$

Proof. This is a nice application of the idea that subspaces do not have inverses. For example, we could have  $V$  as  $\mathbb{R}^2$ ,  $V_1 = \{(x, 0) : x \in \mathbb{R}\}$ , and  $V_2 = \{(0, y) : y \in \mathbb{R}\}$ , and  $U = V$ . These both add to  $V$ , but they are not equal.  $\square$

21 Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in F\}$$

Find a subspace  $W$  of  $\mathbf{F}^5$  such that  $\mathbf{F}^5 = U \oplus W$ .

Proof. We have 2 variables, so we should have 3 variables in the other one to counteract this. The first 2 terms of  $U$  are just  $x$  and  $y$ , which is great because now they cannot vary. We can choose

$$W = \{(0, 0, p, q, r) \in \mathbf{F}^5 : p, q, r \in \mathbf{F}\}$$

And then any vector of the form  $(a, b, c, d, e)$  using

$$\begin{aligned} x &= a \\ y &= b \\ p &= c - (x + y) = c - a - b \\ q &= d - (x - y) = c - a + b \\ r &= e - (2x) = e - 2a \end{aligned}$$

$\square$

23 Prove or give a counterexample: If  $V_1, V_2, U$  are subspaces of  $V$  such that

$$V = V_1 \oplus U \text{ and } V = V_2 \oplus U$$

then  $V_1 = V_2$ .

Proof. This is untrue. Choose  $V = \mathbb{R}^2$ . Let  $U = \{(x, 0) : x \in \mathbb{R}\}$ . Then choose  $V_1 = \{(0, b) : b \in \mathbb{R}\}$  but  $V_2 = \{(a, a) : a \in \mathbb{R}\}$ .

These are different subspaces, but it can be shown that their direct sum with  $U$  is both  $V$ .  $\square$

24 A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called even if

$$f(-x) = f(x)$$

for all  $x \in \mathbb{R}$ . A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called odd if

$$f(-x) = -f(x)$$

for all  $x \in \mathbb{R}$ . Let  $V_e$  denote the set of real-valued even functions on  $\mathbb{R}$  and let  $V_o$  denote the set of real-valued odd functions on  $\mathbb{R}$ . Show that

$$\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$$

Proof. Given some function  $f : \mathbb{R} \rightarrow \mathbb{R}$  we can construct it by adding the following functions:

We have  $g \in V_e$ ,

$$g(x) = \frac{f(x) + f(-x)}{2}$$

It is clearly even by symmetry.

We can then choose  $h \in V_o$ ,

$$h(x) = \frac{f(x) - f(-x)}{2}$$

This is clearly odd.

Now we must show that these are the only two functions that create  $f$ . Given some  $g$  and  $h$ , we must have  $f(x) - g(x) = h(x)$ . This means that

$$\begin{aligned} f(-x) - g(-x) &= -h(x) \\ f(-x) - g(x) &= -h(x) \\ f(-x) &= g(x) - h(x) \\ f(x) &= g(x) + h(x) \\ f(-x) + f(x) &= 2g(x) \\ g(x) &= \frac{f(-x) + f(x)}{2} \end{aligned}$$

$h$  comes naturally afterward.

□

## 2 Finite Dimensional Vector Spaces

We say goodbye to our infinite dimensional vector spaces (noooo not  $\mathbb{R}^{\mathbb{R}}$ ) and we now focus on vector spaces which can actually be spanned by a list of vectors. This unit is... harder than the first one definitely, and it feels more like “linear algebra” than just “abstract algebra.”

### 2.A Span and Linear Dependence

Not too bad.

- 1 Find a list of four distinct vectors in  $\mathbf{F}^3$  whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}$$

Proof. Actually, I only need 2 vectors for this.

$$v_1 = (1, -1, 0) \text{ and } v_2 = (0, 1, -1)$$

With these, I can construct any vector that satisfies the properties. I can include some other ones but I am lazy.

□

- 3 Suppose  $v_1, \dots, v_m$  is a list of vectors in  $V$ . For  $k \in \{1, \dots, m\}$ , let

$$w_k = v_1 + \dots + v_k$$

Show that  $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$ .

Proof. Given an arbitrary linear combination of  $w_k$ ,

$$\sum_{k=1}^m a_k w_k = \sum_{k=1}^m \left( \sum_{i=1}^k a_i \right) v_k$$

For a given  $v_k$ , we can construct it from a linear combination of  $w$  as  $v_k = w_k - w_{k-1}$  where  $v_1 = w_1$ . Thus, since we can go both ways, the spans are equal.

□

- 5 Find a number  $t$  such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in  $\mathbf{R}^3$

Proof. If we can find some

$$a(3, 1, 4) + b(2, -3, 5) = (5, 9, t)$$

then our list of vectors is not linearly independent. This turns into these equations:

$$3a + 2b = 5$$

$$a - 3b = 9$$

$$4a + 5b = t$$

We solve, and we get that  $a = 3, b = -2, t = 2$ , so it is not linearly independent when  $t = 2$ .  $\square$

- 7 (a) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{R}$ , then the list  $1 + i, 1 - i$  is linearly independent.

Proof. Clearly there is no positive multiple of  $1 + i$  that creates  $1 - i$  you dumbass. The real and imaginary parts will always have the same sign.  $\square$

- (b) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{C}$ , then the list  $1 + i, 1 - i$  is linearly dependent.

Proof. We can multiple  $1 + i$  by

$$\begin{aligned} \frac{1 - i}{1 + i} &= \frac{(1 - i)(1 - i)}{(1 + i)(1 - i)} \\ &= \frac{-2i}{2} \\ &= -i \end{aligned}$$

to get  $1 - i$ , therefore its linearly dependent.  $\square$

- 9 Prove or give a counterexample: If  $v_1, v_2, \dots, v_m$  is a linearly independent list of vectors in  $V$ , then

$$5v_1 - 4v_2, v_2, v_3, \dots, v_m$$

is linearly independent.

Proof. Suppose that  $5v_1 - 4v_2, v_2, v_3, \dots, v_m$  was linearly dependent. Then,

$$a_1(5v_1 - 4v_2) + a_2v_2 + \dots + a_mv_m = 5a_1v_1 + (a_2 - 4a_1)v_2 + \dots + a_mv_m = 0$$

and we have a contradiction.  $\square$

- 11 Prove or give a counterexample: If  $v_1, \dots, v_m$  and  $w_1, \dots, w_m$  are linearly independent lists of vectors in  $V$ , then the list  $v_1 + w_1, \dots, v_m + w_m$  is linearly independent.

Proof. We can construct a counterexample. Let us use  $\mathbb{R}^1$  here. 3 is linearly independent, and so is  $-3$ , but  $3 - 3 = 0$ , which is not linearly independent.  $\square$

- 13 Suppose  $v_1, \dots, v_m$  is linearly independent in  $V$  and  $w \in V$ . Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m).$$

Proof. We prove the contrapositive in both directions.

Suppose that  $w \in \text{span}(v_1, \dots, v_m)$ . Then, there is some linear combination of  $v_1, \dots, v_m$  and so  $v_1, \dots, v_m, w$  is linearly dependent.

Suppose that  $v_1, \dots, v_m, w$  is linearly dependent. Then, by the linear dependence lemma, there exists some element in this list that is in the span of the previous ones. Because all lists  $v_1, \dots, v_k, k \leq m$  are linearly independent,  $w \in \text{span}(v_1, \dots, v_m)$ .  $\square$

- 15 Explain why there does not exist a list of six polynomials that is linearly independent in  $\mathcal{P}_4(\mathbf{F})$ .

Proof. We have a list of 5 polynomials that spans  $\mathcal{P}_4(\mathbf{F})$ . The length of a linearly independent list must be less than or equal to the length of this spanning list, so there does not exist a list of six polynomials that is linearly independent in  $\mathcal{P}_4(\mathbf{F})$ .  $\square$

- 17 Prove that  $V$  is infinite-dimensional if and only if there is a sequence  $v_1, v_2, \dots$  of vectors in  $V$  such that  $v_1, \dots, v_m$  is linearly independent for every positive integer  $m$ .

Proof. Because the length of the span of a vector space must be greater or equal to any given list of linearly independent vectors (Replacement theorem), this sequence means that there is no finite list of vectors that spans  $V$ .

If  $V$  is infinite-dimensional, we can construct such a sequence  $v_1, v_2, \dots$ . Begin by choosing an arbitrary vector  $v_1 \neq 0$ . Then, for each  $k \geq 2$ , we know that

$$\text{span}(v_1, \dots, v_{k-1}) \neq V$$

So there exists some  $v_k \notin \text{span}(v_1, \dots, v_{k-1})$ . Continue for an arbitrary number of steps.  $\square$

- 19 Prove that the real vector space of all continuous real-valued functions on the interval  $[0, 1]$  is infinite-dimensional.

Proof. I can construct a sequence of functions that are all linearly independent. In fact, I can use

$$1, x, x^2, x^3 \dots$$

Since each polynomial is uniquely determined by its coefficients, these are linearly independent, and they are also clearly continuous.  $\square$

- 20 Suppose  $p_0, p_1, \dots, p_m$  are polynomials in  $\mathcal{P}_m(\mathbf{F})$  such that  $p_k(2) = 0$  for each  $k \in \{0, \dots, m\}$ . Prove that  $p_0, p_1, \dots, p_m$  is not linearly independent in  $\mathcal{P}_m(\mathbf{F})$ .

Proof. For  $m = 0$ ,  $p_0 = 0$ , which isn't linearly independent.

For  $m > 0$ , if  $p_k = 0$  for some  $k \in \{0, \dots, m\}$ , then our set is already not linearly independent.

Suppose  $p_k \neq 0$  for all  $k \in \{0, \dots, m\}$ . Then, because  $p_k(2) = 0$ , all  $p_k$  has a factor of  $(z - 2)$ . When finding a linear combination that equals zero, we can factor out all by  $(z - 2)$ . That is

$$a_1 p_1 + a_2 p_2 + \dots + a_m p_m = 0 \iff (z - 2) \left( a_1 \frac{p_1}{z - 2} + \dots + a_m \frac{p_m}{z - 2} \right) = 0$$

Now, we have a set of polynomials of degree  $m - 1$ , but we have  $m + 1$  terms. For  $\mathcal{F}_{m-1}(\mathbf{F})$  there is a list of  $m$  vectors that span it. Thus, the length of a linearly independent list of vectors must be less than or equal to  $m$ . We have  $m + 1$  terms in our list, so we can determine that our list is linearly dependent.  $\square$

## 2.B Bases

This is more or less a merging of the concepts of span and linear independence and showing their relationships. Pretty easy still.

1. Find all vector spaces that have exactly one basis.

Proof. We have the vector space of size 1 and dimension 0,  $\{0\}$ . Its only basis is the empty set. In one dimension, we have the basis for  $\mathbb{F}_2^1$ , the vector space consisting of only one dimension of the field of size 2, which has the basis 1. If the field had more elements than 2, then you would be able to obtain another basis by multiplying one of the elements by some non-zero scalar. If the vector space had more than 1 dimension, we could choose other bases, such as  $\{1, 1\}, \{0, 1\}$  for  $\mathbb{F}_2^2$ .  $\square$

- 3 (a) Let  $U$  be the subspace of  $\mathbb{R}^5$  defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 = 3x_2, x_3 = 7x_4\}.$$

Find a basis of  $U$ .

Proof.  $x_1$  and  $x_4$  are useless, so we can just use the standard basis vectors for  $\mathbb{R}^3$  except instead using filled in values for  $x_1$  and  $x_4$ . For example,

$$\{3, 1, 0, 0, 0\}$$

$$\{0, 0, 7, 1, 0\}$$

$$\{0, 0, 0, 0, 1\}$$

□

- 5 Suppose  $V$  is finite-dimensional and  $U, W$  are subspaces of  $V$  such that  $V = U + W$ . Prove that there exists a basis of  $V$  consisting of vectors in  $U \cup W$ .

Proof. If there is a spanning list of vectors in  $U \cup W$ , then there is also a basis in  $U \cup W$  (since every spanning list can be reduced to a basis).

We can thus attach a basis of  $U$  to a basis in  $W$  as our spanning list. This list spans  $V$  because  $V = U + W$  implies for every  $v \in V, \exists u \in U, w \in W$  s.t.  $v = u + w$ , and since we can create any  $u + w$  in our new basis, we are done. □

- 7 Suppose  $v_1, v_2, v_3, v_4$  is a basis of  $V$ . Prove that  $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$  is also a basis of  $V$ .

Proof. Clearly, our given vectors are in the span of  $v_1, \dots, v_4$ .

We will show that you can represent  $v_1, \dots, v_4$  as linear combinations of our given vectors (which will thus show that they have the same span). After we derive one vector, we can use it in the formula of other vectors (implicitly applying a substitution to keep it in our vector set)

$$v_4 = v_4$$

$$v_3 = (v_3 + v_4) - v_4$$

$$v_2 = (v_2 + v_3) - v_3$$

$$v_1 = (v_1 + v_2) - v_2$$

These vectors must also be linearly independent. Suppose that they were not linearly independent. Then, we could remove a vector without changing its span. Now, we have a span of vectors that is smaller than a linearly independent list of vectors. Contradiction! □

- 8 Prove or give a counterexample: If  $v_1, v_2, v_3, v_4$  is a basis of  $V$  and  $U$  is a subspace of  $V$  such that  $v_1, v_2 \in U$  and  $v_3 \notin U$  and  $v_4 \notin U$ , then  $v_1, v_2$  is a basis of  $U$ .

Proof. Nope! Suppose we are working in  $\mathbb{R}^4$  and our basis is

$$(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 1), (0, 0, 0, 1)$$

We could have a subspace  $U$  that admits basis

$$(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0)$$

□

## 2.C Dimension

THESE PROBLEMS GOT A PUNCH TO EM!!! Well, mainly I struggled on 8. The rest were fine.

1. Show that the subspaces of  $\mathbb{R}^2$  are precisely  $\{0\}$ , all lines in  $\mathbb{R}^2$  containing the origin, and  $\mathbb{R}^2$ .

Proof. We can split this up by the number of dimensions that the basis of the subspace contains.

- (a) 0 dimensions. The only 0 dimensional vector is  $\{0\}$ .
- (b) 1 dimension. This is a single vector, and it will always produce a line after scaling in  $\mathbb{R}^2$ .
- (c) 2 dimensions. Because the 2 dimensions are linearly independent, we know that it spans all of  $\mathbb{R}^2$ .

□

- 3 (a) Let  $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(6) = 0\}$ . Find a basis of  $U$ .

Proof. If  $p(6) = 0$ , then all polynomials in  $U$  will have a factor of  $(x-6)$  (including 0 by technicality). Thus, we can choose the basis

$$\begin{aligned} &(x-6) \\ &(x-6)(x) \\ &(x-6)(x^2)(x-6)(x^3) \end{aligned}$$

□

- (b) Extend the basis in (a) to a basis of  $\mathcal{P}_4(\mathbb{F})$ .

Proof. We can simply add the constant, 1. This is now a linearly independent list which has length  $\dim \mathcal{P}_4(\mathbb{F})$ , so it is a basis. □

- (c) Find a subspace  $W$  of  $\mathcal{P}_4(\mathbb{F})$  such that  $\mathcal{P}_4(\mathbb{F}) = U \oplus W$ .

Proof. It is just the subspace given by the span of the additional vector, in this case

$$W = \{a : a \in \mathbb{F}\}$$

□

- 5 (a) Let  $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$ . Find a basis of  $U$ .

Proof. We have a fixed point, which should reduce the dimension by 1 but not in an entirely clear way...

We know there must not be a polynomial of degree exactly 1 because that would make it impossible to satisfy the constraints, so we can simply choose polynomials that satisfy the property. Such as...

$$\begin{aligned} &1 \\ &(x-2)(x-5) \\ &(x-2)(x-5)x \\ &(x-2)(x-5)x^2 \end{aligned}$$

These 3 vectors are clearly linearly independent and are in  $U$ . We know that it spans  $U$  because it has less than 4 dimensions (otherwise it would equal to  $\mathcal{P}_4(\mathbb{F})$ ) and since this is a linearly independent list of 3 it can't be any lower. □

- (b) Extend the basis in (a) to a basis of  $\mathcal{P}_4(\mathbb{F})$ .

Proof. We can plop in  $x$  now. □

- (c) Find a subspace  $W$  of  $\mathcal{P}_4(\mathbb{F})$  such that  $\mathcal{P}_4(\mathbb{F}) = U \oplus W$ .

Proof. That would be  $W = \{ax : a \in \mathbb{F}\}$ . □

- 7 (a) Let  $U = \{p \in \mathcal{P}_4(\mathbb{R}) : \int_{-1}^1 p = 0\}$ . Find a basis of  $U$ .

Proof. We can create a list of 4 linearly independent vectors. 3 is the maximum basis size for a (strict) subspace of  $\mathcal{P}_4(\mathbb{F})$  so we are sure it spans the whole subspace.

$$\begin{aligned} &x \\ &x^3 \\ &x^2 - \frac{1}{3} \\ &x^4 - \frac{1}{5} \end{aligned}$$

□

(b) Extend the basis in (a) to a basis of  $\mathcal{P}_4(\mathbb{R})$ .

Proof. We can add just one more linearly independent element, say, 1.  $\square$

(c) Find a subspace  $W$  of  $\mathcal{P}_4(\mathbb{R})$  such that  $\mathcal{P}_4(\mathbb{R}) = U \oplus W$ .

Proof.

$$W = \{a : a \in \mathbb{F}\}$$

$\square$

8 Suppose  $v_1, \dots, v_m$  is linearly independent in  $V$  and  $w \in V$ . Prove that

$$\dim \text{span}(v_1 + w, \dots, v_m + w) \geq m - 1$$

Proof. We can subtract the first vector from the rest to obtain that

$$v_2 - v_1, \dots, v_m - v_1 \in \text{span}(v_1 + w, \dots, v_m + w)$$

Therefore,

$$\text{span}(v_2 - v_1, \dots, v_m - v_1) \leq \text{span}(v_1 + w, \dots, v_m + w)$$

Because

$$\text{span}(v_1, \dots, v_m) = \text{span}(v_1, v_2 - v_1, \dots, v_m - v_1) = m$$

We have that

$$\text{span}(v_1 + w, \dots, v_m + w) \geq \text{span}(v_2 - v_1, \dots, v_m - v_1) = m - 1$$

$\square$

9 Suppose  $m$  is a positive integer and  $p_0, p_1, \dots, p_m \in \mathcal{P}(\mathbb{F})$  are such that each  $p_k$  has degree  $k$ . Prove that

$$p_0, p_1, \dots, p_m$$

is a basis of  $\mathcal{P}_m(\mathbb{F})$ .

Proof. We prove it by induction. First,  $\mathcal{P}_0(\mathbb{F})$  is just  $\mathbb{F}$ . Any non-zero value in  $\mathbb{F}$  is a basis for  $\mathbb{F}$ .

Suppose that  $p_0, \dots, p_i$  was a basis for  $\mathcal{P}_i(\mathbb{F})$ . Then, we introduce arbitrary  $i + 1$ th degree polynomial  $p_{i+1}$ . Given some arbitrary polynomial in  $\mathcal{P}_{i+1}(\mathbb{F})$ ,  $f(x)$ , we can set the coefficient of  $p_{i+1}$  to be the coefficient of the  $x^{i+1}$  term of  $f$ . Then, what remains after subtracting the  $p_{i+1}$  term out is a polynomial in  $\mathcal{P}_i(\mathbb{F})$ . Because we know that

$p_0, \dots, p_i$  is a basis of  $\mathcal{F}_i(\mathbb{F})$ , we know we can produce the rest of the polynomial through a linear combination of  $p_0, \dots, p_i$ . Thus,  $p_0, \dots, p_{i+1}$  spans  $\mathcal{P}_{i+1}(\mathbb{F})$ . Because  $\mathcal{P}_{i+1}(\mathbb{F})$  has dimension  $i + 2$  and we have a list of  $i + 2$  vectors that span it, we know that this list is also linearly independent, and thus we know that  $p_0, \dots, p_{i+1}$  is a basis for  $\mathcal{P}_{i+1}(\mathbb{F})$ .

Therefore by induction for any  $m \in \mathbb{N}$ ,  $p_0, \dots, p_m$  is a basis for  $\mathcal{P}_m(\mathbb{F})$ .  $\square$

- 10 Suppose  $m$  is a positive integer. For  $0 \leq k \leq m$ , let

$$p_k(x) = x^k(1 - x)^{m-k}.$$

Show that  $p_0, \dots, p_m$  is a basis of  $\mathcal{P}_m(\mathbb{F})$ .

Proof. We can prove this by observing that these polynomials are actually quite similar to the ones from last exercise. For example,  $p_m$  is a polynomial that contains non-zero terms for degree  $m$  only.  $p_{m-1}$  has non-zero terms for degree terms for degrees  $m$  and  $m - 1$ . Thus, we can use the same argument as the previous exercise, using induction, but this time in reverse.  $\square$

- 11 Suppose  $U$  and  $W$  are both four-dimensional subspaces of  $\mathbb{C}^6$ . Prove that there exist two vectors in  $U \cap W$  such that neither of these vectors is a scalar multiple of the other.

Proof. The dimension of  $U + W$  is at most the dimension of the space itself (6). Therefore, we have that

$$6 \geq \dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

$$6 \geq 4 + 4 - \dim(U \cap W)$$

$$2 \leq \dim(U \cap W)$$

Now, we can use 2 basis vectors for  $\dim(U \cap W)$  as our pair of linearly independent vectors.  $\square$

- 15 Suppose  $V$  is finite-dimensional and  $V_1, V_2, V_3$  are subspaces of  $V$  with  $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$ . Prove that  $V_1 \cap V_2 \cap V_3 \neq \{0\}$ .

Proof.

$$\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$$

$$\geq \dim V + \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$$

$$\dim V_3 > \dim V - \dim(V_1 \cap V_2)$$

$$\geq \dim(V_1 \cap V_2) + \dim(V_3) - \dim(V_1 \cap V_2 \cap V_3) - \dim(V_1 \cap V_2)$$

$$\dim(V_1 \cap V_2 \cap V_3) > 0$$

$\square$

- 17 Suppose that  $V_1, \dots, V_m$  are finite-dimensional subspaces of  $V$ . Prove that  $V_1 + \dots + V_m$  is finite-dimensional and

$$\dim(V_1 + \dots + V_m) \leq \dim V_1 + \dots + \dim V_m.$$

Proof. We prove by induction on  $m$ . For  $m = 1$ , the statement is trivial.

Assume the result holds for  $m - 1$  subspaces. By the inductive hypothesis,  $V_1 + \dots + V_{m-1}$  is finite-dimensional. Then  $V_1 + \dots + V_m = (V_1 + \dots + V_{m-1}) + V_m$  is a sum of two finite-dimensional subspaces. By the dimension formula for two subspaces,

$$\dim((V_1 + \dots + V_{m-1}) + V_m) \leq \dim(V_1 + \dots + V_{m-1}) + \dim V_m.$$

By the inductive hypothesis,  $\dim(V_1 + \dots + V_{m-1}) \leq \dim V_1 + \dots + \dim V_{m-1}$ . Therefore,

$$\dim(V_1 + \dots + V_m) \leq \dim V_1 + \dots + \dim V_{m-1} + \dim V_m.$$

Thus by induction, the result holds for all positive integers  $m$ .  $\square$

- 19 Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if  $V_1, V_2, V_3$  are subspaces of a finite-dimensional vector space, then

$$\begin{aligned} \dim(V_1 + V_2 + V_3) = & \\ & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ & + \dim(V_1 \cap V_2 \cap V_3). \end{aligned}$$

Then either prove the formula above or give a counterexample.

Proof. This statement is FALSE. Consider 3 different lines in  $\mathbb{R}^2$ . We would have  $2 = 1 + 1 + 1 - 0 - 0 - 0 + 0 = 3!$   $\square$

- 20 Prove that if  $V_1, V_2$ , and  $V_3$  are subspaces of a finite-dimensional vector space, then

$$\begin{aligned} \dim(V_1 + V_2 + V_3) = & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ & + \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3} \end{aligned}$$

Proof. This is really 3 formulas squished into one.

We have

$$\begin{aligned} \dim((V_1 + V_2) + V_3) &= \dim(V_1 + V_2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \end{aligned}$$

We can apply this again for other permutations to obtain three formulas. Summing them together and dividing by three will give the desired formula. I would write it up but I really gotta get to chapter 3 now.  $\square$

### 3 Linear Maps

A linear map from  $V$  to  $W$  is a function  $T : V \rightarrow W$  with the following properties.

- additivity  $T(u + v) = Tu + Tv$  for all  $u, v \in V$ .
- homogeneity  $T(\lambda v) = \lambda(Tv)$  for all  $\lambda \in \mathbf{F}$  and all  $v \in V$ .

This chapter is all about em!

#### 3.A Vector Space of Linear Maps

Very light on the notes side, probably because this is such a critical new topic. Problems 11 and 17 were basically impossible, I attempted them but it was not possible for me to solve them unfortunately.

1. Suppose  $b, c \in \mathbb{R}$ . Define  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  by  $T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz)$ . Show that  $T$  is linear if and only if  $b = c = 0$ .

Proof. Suppose that  $b \neq 0$ . Then,  $T(0) = (b, 0) \neq 0$ , so its not a linear map.

Suppose that  $c \neq 0$ . Then, look at  $T((1, 1, 1) + (1, 1, 1)) = T((2, 2, 2)) = (2, 12 + 8c) \neq T(1, 1, 1) + T(1, 1, 1) = (2, 12 + c)$ .

Now let's show that when  $b = 0, c = 0$ ,  $T$  is a linear map. Suppose we have  $a, b, \in \mathbb{R}^3$ .

$$\begin{aligned} T(a + b) &= (2a_1 + 2b_1 - 4a_2 - 4b_2 + 3a_3 + 3b_3, 6a_1 + 6b_1) \\ &= (2a_1 - 4a_2 + 3a_3, 6a_1) + (2b_1 - 4b_2 + 3b_3, 6b_1) \\ &= T(a) + T(b) \end{aligned}$$

□

2. Suppose  $b, c \in \mathbb{R}$ . Define  $T : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2$  by

$$Tp = \left( 3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^2 x^3 p(x) dx + c \sin p(0) \right)$$

Show that  $T$  is linear if and only if  $b = c = 0$ .

Proof. This thing is so bad honestly its easier to prove that the sum of a linear map and a non-linear map is non-linear.

Suppose we have functions  $f \in \mathcal{L}(U, W)$  and  $g : U \rightarrow W$  (in this case,  $U$  and  $W$  are just sets who happen to be interpretable as vector spaces). If  $f + g$  was a linear map, so would  $f + g - f$ , which clearly isn't.

Then, we know that linear maps are preserved under addition, scalar multiplication, and function composition, so everything except the terms with  $b$  and  $c$  are cleared.

$bp(1)p(2)$  is not a linear map (when  $b \neq 0$ ) because it does not preserve scalar multiplication. Multiplying  $p$  by a constant multiplies  $bp(1)p(2)$  by its square.  $c \sin p(0)$  clearly violates both addition and multiplication (sin isn't linear!). So,  $b = c = 0$ . Since everything else is a linear map,  $T$  must be too.  $\square$

3. Suppose that  $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ . Show that there exist scalars  $A_{j,k} \in \mathbb{F}$  for  $j = 1, \dots, m$  and  $k = 1, \dots, n$  such that for every  $(x_1, \dots, x_n) \in \mathbb{F}^n$ ,

$$T(x_1, \dots, x_n) = \left( \sum_{k=1}^n A_{1,k}x_k, \dots, \sum_{k=1}^n A_{m,k}x_k \right).$$

Proof. Consider the standard basis for  $\mathbb{F}^n$ . The Linear Map Lemma states that a linear map is entirely determined by where it sends its basis vectors. Let

$$Te_i = (A_{1,i}, \dots, A_{m,i})$$

Then,

$$\begin{aligned} T(x_1, \dots, x_n) &= Tx_1 + \dots + Tx_n \\ &= (A_{1,1}, \dots, A_{m,1}) + \dots + (A_{1,n}, \dots, A_{m,n}) \\ &= \left( \sum_{k=1}^n A_{1,k}x_k, \dots, \sum_{k=1}^n A_{m,k}x_k \right) \end{aligned}$$

$\square$

- 5 Prove that  $\mathcal{L}(V, W)$  is a vector space, as was asserted in 3.6.

Proof. We are given the following definitions for addition and scalar multiplication:

Given  $S, T, \in \mathcal{L}(V, W)$  and  $\lambda \in \mathbb{F}$ ,

$$\begin{aligned} (S + T)(v) &= Sv + Tv \\ (\lambda T)(v) &= \lambda(Tv) \end{aligned}$$

Just like the proofs for vector spaces themselves, we can strap our properties to the existing properties of our ring.

(a) Commutativity

$$\begin{aligned} (S + T)(v) &= Sv + Tv \\ &= Tv + Sv \text{ (Commutativity of } W) \\ &= (T + S)(v) \end{aligned}$$

(b) Associativity

$$\begin{aligned}((S + T) + H)(v) &= (S + T)(v) + H(v) \\&= Sv + Tv + Hv \\&= Sv + (Tv + Hv) \\&= (S + (T + H))(v)\end{aligned}$$

(c) Additive Inverse

Define  $(-T)(v) = -Tv$ .

$$(T + (-T))(v) = Tv - Tv = 0$$

(d) Multiplicative Identity

Our identity element is 1. This is trivially the identity, shush.

(e) Distributive property

$$\begin{aligned}(\lambda(S + T))(v) &= \lambda(S + T)(v) \\&= \lambda(Sv + Tv) \\&= \lambda Sv + \lambda Tv &= (\lambda S + \lambda T)(v)\end{aligned}$$

□

8 Give an example of a function  $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}$  such that

$$\varphi(av) = a\varphi(v)$$

for all  $a \in \mathbb{R}$  and all  $v \in \mathbb{R}^2$ , but  $\varphi$  is not linear.

Proof. Maybe something like

$$f(a, b) = \begin{cases} a & a = b \\ 0 & a \neq b \end{cases}$$

would work.

□

9 Give an example of a function  $\varphi: \mathbb{C} \rightarrow \mathbb{C}$  such that

$$\varphi(w + z) = \varphi(w) + \varphi(z)$$

for all  $w, z \in \mathbb{C}$  but  $\varphi$  is not linear. (Here  $\mathbb{C}$  is thought of as a complex vector space.)

Proof. Something like  $\varphi(z) = \bar{z}$  would work. It satisfies additivity but

$$i\overline{1+i} \neq \overline{i(1+i)}$$

□

- 10 Prove or give a counterexample: If  $q \in \mathcal{P}(\mathbb{R})$  and  $T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$  is defined by  $Tp = q \circ p$ , then  $T$  is a linear map.

Proof. Not true in general, since something like  $x^2$  violates the additivity constraint.  $((1+1)^2 \neq 2^2)$  □

- 11 Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ . Prove that  $T$  is a scalar multiple of the identity if and only if  $ST = TS$  for every  $S \in \mathcal{L}(V)$ .

Proof. I cannot seem to find a good solution to this exercise online, so it is skipped. Maybe I will come back to it. □

- 12 Suppose  $U$  is a subspace of  $V$  with  $U \neq V$ . Suppose  $S \in \mathcal{L}(U, W)$  and  $S \neq 0$  (which means that  $Su \neq 0$  for some  $u \in U$ ). Define  $T: V \rightarrow W$  by

$$Tv = \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U. \end{cases}$$

Prove that  $T$  is not a linear map on  $V$ .

Proof. Suppose there was some  $u \in U$  where  $Su \neq 0$ . Then, we have some  $v \in V, v \notin U$ .  $u + v \notin U$ , so  $T(u + v) = 0$ . But also  $T(u + v) = Tu + Tv = Tu \neq 0$ . Contradiction! □

- 15 Suppose  $v_1, \dots, v_m$  is a linearly dependent list of vectors in  $V$ . Suppose also that  $W \neq \{0\}$ . Prove that there exist  $w_1, \dots, w_m \in W$  such that no  $T \in \mathcal{L}(V, W)$  satisfies  $Tv_k = w_k$  for each  $k = 1, \dots, m$ .

Proof. We have some linear combination,  $a_i$  not all 0, such that

$$a_1v_1 + \dots + a_mv_m = 0$$

Let  $a_i$  be the first non-zero index. We let  $w_i$  be an arbitrary non-zero vector in  $W$  and all other  $j \neq i, w_j = 0$ .

Then, we have that

$$T(a_1v_1 + \dots + a_mv_m) = 0$$

But also

$$\begin{aligned} T(a_1v_1 + \cdots + a_mv_m) &= a_1T(v_1) + \cdots + a_mT(v_m) \\ &= a_1w_1 + \cdots + a_mw_m \\ &= a_iw_i \\ &\neq 0 \end{aligned}$$

So no such  $T$  exists. □

- 17 Suppose  $V$  is finite-dimensional. Show that the only two-sided ideals of  $\mathcal{L}(V)$  are  $\{0\}$  and  $\mathcal{L}(V)$ . A subspace  $\mathcal{E}$  of  $\mathcal{L}(V)$  is called a two-sided ideal of  $\mathcal{L}(V)$  if  $TE \in \mathcal{E}$  and  $ET \in \mathcal{E}$  for all  $E \in \mathcal{E}$  and all  $T \in \mathcal{L}(V)$ .

Proof. This problem is cooked. □

### 3.B Null Spaces and Ranges

- 1 Give an example of a linear map  $T$  with  $\dim \text{null } T = 3$  and  $\dim \text{range } T = 2$ .

Proof.

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

So  $V$  is 5 dimensional. Lets use  $\mathbb{F}^5$ . Then, we can set

$$T(a_1, \dots, a_5) = a_1, a_2, 0, 0, 0$$

The range is clearly 2 dimensional, and the null space is clearly 3 dimensional. □

- 3 Suppose  $v_1, \dots, v_m$  is a list of vectors in  $V$ . Define  $T \in \mathcal{L}(\mathbb{F}^m, V)$  by

$$T(z_1, \dots, z_m) = z_1v_1 + \cdots + z_mv_m.$$

- (a) What property of  $T$  corresponds to  $v_1, \dots, v_m$  spanning  $V$ ?

Proof.  $\text{range } T = V$  corresponds with spanning  $V$ . □

- (b) What property of  $T$  corresponds to the list  $v_1, \dots, v_m$  being linearly independent?

Proof.  $\text{null } T = \{0\}$  corresponds with linear independence. □

- 7 Suppose  $V$  and  $W$  are finite-dimensional with  $2 \leq \dim V \leq \dim W$ . Show that

$$\{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$$

is not a subspace of  $\mathcal{L}(V, W)$ .

Proof. Because  $T$  is not injective, we know that  $\dim \text{null } T \neq 0$ . Thus, given  $T_1$  where  $T_1 \neq 0$ , let us define some  $T_2$  as

$$T_2(v) = \begin{cases} \text{non-zero value} & T_1(v) = 0 \\ 0 & T_1(v) \neq 0 \end{cases}$$

The specific non-zero value can be defined using basis stuff, but the details are not too important here. What is important is that null and range are both subspaces, so by swapping which is 0 and which is non-zero, we will keep these subspaces.

range  $T_2 = \text{null } T_1$ .  $T_2$  will also be not injective because we swap range and null, and we specify that the range of  $T_1$  must be non-zero. Now the null space of  $(T_1 + T_2) = \{0\}$ , which is injective, so we have our proof that it is not closed under addition (and therefore not a subspace).  $\square$

note Big gap here, on account of the problems in between being too easy.

- 18 Suppose  $V$  and  $W$  are finite-dimensional and that  $U$  is a subspace of  $V$ . Prove that there exists  $T \in \mathcal{L}(V, W)$  such that  $T = U$  if and only if

$$\dim U \geq \dim V - \dim W.$$

Proof. Suppose there was a linear map  $T$  with  $\text{null } T = U$ . Then,

$$\begin{aligned} \dim V &= \dim \text{null } T + \dim \text{range } T \\ &= \dim U + \dim \text{range } T \\ &\leq \dim U + \dim W \\ \dim U &\geq \dim V - \dim W \end{aligned}$$

And the converse is also true because trivial (we can construct using basis stuff)  $\square$

- 19 Suppose  $W$  is finite-dimensional and  $T \in \mathcal{L}(V, W)$ . Prove that  $T$  is injective if and only if there exists  $S \in \mathcal{L}(W, V)$  such that  $ST$  is the identity operator on  $V$ .

Proof. Suppose that  $ST$  was the identity operator. Then, if  $v_1, v_2 \in V, v_1 \neq v_2$  and  $Tv_1 = Tv_2$ , then  $STv_1 = STv_2$ , which implies  $v_1 = v_2$ , contradiction.

Suppose that  $T$  is injective. Suppose  $V$  has basis vectors  $v_1, \dots, v_n$  and  $W$  has basis vectors  $w_1, \dots, w_n$  for some subspace of  $W$  (we know that  $\dim V = \dim W$  because otherwise we would not have an injection). Also,  $Tv_i = w_i$  (if it did not map to the basis, it would not be injective, proof is trivial but you use range for it). Then we could also define  $Sw_i = v_i$ , which is a linear map (trivially), and we are done.  $\square$

- 21 Suppose  $V$  is finite-dimensional,  $T \in \mathcal{L}(V, W)$ , and  $U$  is a subspace of  $W$ . Prove that  $\{v \in V : Tv \in U\}$  is a subspace of  $V$  and

$$\dim\{v \in V : Tv \in U\} = \dim \text{null } T + \dim(U \cap \text{range } T).$$

Proof. First,  $\{v \in V : Tv \in U\}$  is a subspace of  $V$ . Let's name it  $X$ . Let  $x_1, x_2 \in X$ , then  $T(x_1 + x_2) = Tx_1 + Tx_2 \in U$ . Same works with scalar multiples.

Next, let's define  $T_1 \in \mathcal{L}(X, W)$ ,  $T_1v = Tv$  for all  $v \in X$ . By the fundamental theorem of linear maps,

$$\dim X = \dim \text{null } X + \dim \text{range } X$$

$$\dim X = \dim \text{null } T + \dim \text{range } X$$

(we know this because any  $v$  that maps to 0 is in  $X$ )

$$\dim\{v \in V : Tv \in U\} = \dim \text{null } T + \dim(U \cap \text{range } T)$$

□

- 24 Suppose  $\dim V = 5$  and  $S, T \in \mathcal{L}(V)$  are such that  $ST = 0$ .

- (a) Prove that  $\dim \text{range } TS \leq 2$ .

Proof. Because  $\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T$  and  $\dim \text{null } ST = 5$ , we know that the sum of  $\dim \text{null } S + \dim \text{null } T$  is at least 5. We also know that  $\dim \text{range } TS \leq \min\{\dim \text{range } S, \dim \text{range } T\}$ . The maximum minimum here is clearly 2, so  $\dim \text{range } TS \leq 2$ . □

- (b) Give an example of  $S, T \in \mathcal{L}(\mathbb{F}^5)$  with  $ST = 0$  and  $\dim \text{range } TS = 2$ .

Proof. We can shuffle around the basis vector to get a nice non-communative thing going, too lazy to write it out though. □

- 28 Suppose  $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$  is such that

$$\deg(Dp) = \deg p - 1$$

for every nonconstant polynomial  $p \in \mathcal{P}(\mathbb{R})$ . Prove that  $D$  is surjective.

Proof. First, we prove that constants get mapped to 0.

Let  $D_n$  be  $D$  but in  $\mathcal{L}(\mathcal{P}_n(\mathbb{R}), \mathcal{P}_{n-1}(\mathbb{R}))$ .  $D_1 \neq 0$ , so it ranges all of  $\mathbb{R}$ , which is our base case.

Suppose that  $D_{m-1}$  was surjective. Suppose  $Dm(x^m) = p_m$ . Given arbitrary  $m - 1$ th degree polynomial  $p$ , since  $p_m$  is a  $m - 1$ th degree polynomial, we can find a scalar multiple such that  $[x^{m-1}]p = [x^{m-1}]p_m$ , let's say its  $a$ . Then, we subtract that out of  $p$ . What we are left with is a  $m - 2$ th degree polynomial, and since  $D_{m-1}$  is surjective and  $D_m$  includes  $D_{m-1}$ , have some polynomial that maps to  $p - ap_m$ , and we are done. □